

4.2.17

EE25BTECH11014 - Bhoomika Lokesh

Question:

Find the direction and normal vectors of the line $y = x + 2$

Solution:

The equation of the line can be given by

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

$$\mathbf{n}^T = \begin{pmatrix} 1 & -1 \end{pmatrix} \quad (2)$$

$$c = -2 \quad (3)$$

The equation of line can be expressed as

$$\mathbf{n}^T \mathbf{x} = c \quad (4)$$

where $\mathbf{n}$ is the normal vector of the given line.

The direction vector of the line is perpendicular to $\mathbf{n}$

$$\text{Let } \mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix}$$

We know that,

$$\mathbf{n}^T \mathbf{m} = 0 \quad (5)$$

$$\Rightarrow \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ m \end{pmatrix} = 0 \quad (6)$$

$$1 - m = 0 \quad (7)$$

$$m = 1 \quad (8)$$

$\therefore$ The normal vector of the line is $\mathbf{n}^T = \begin{pmatrix} 1 & -1 \end{pmatrix}$ and the direction vector $\mathbf{m} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

See Fig. 0.1

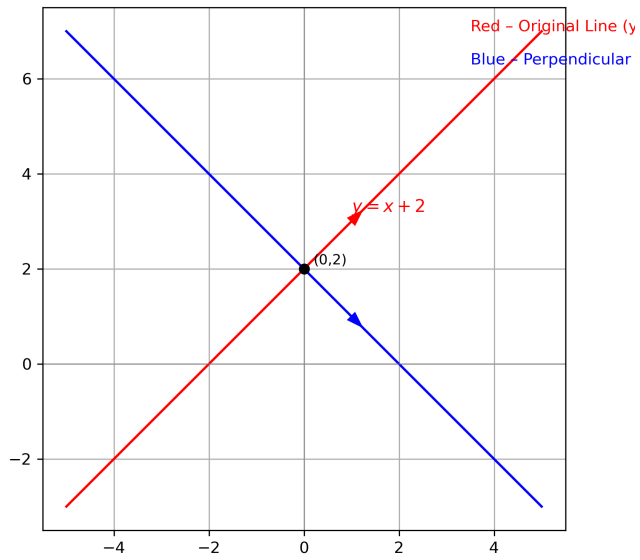


Fig. 0.1: Normal and direction vectors